

Soil Sampling Methods Note

Bulk Density Sampling Justification & ESM Requirements

Purpose of this Note

This methods note addresses two questions that have arisen regarding the soil sampling campaign conducted in September 2025 across five farmer cooperatives in Western Kenya as part of the VM0042 v2.2 Quantification Approach 1 (Measure and Model — RothC) project. Specifically, it clarifies:

- **Part 1:** Whether per-sample bulk density (BD) measurement is required at t=0, or whether stratum-average BD derived from a representative subsample is acceptable.
- **Part 2:** What the Equivalent Soil Mass (ESM) requirement actually demands, when it applies, what additive sampling is needed to satisfy it, and whether the supplemental 30–50 cm BD collection can take place at a different time from the original t=0 SOC sampling.

The short answer to both concerns is that the project is not in a non-conforming position. One targeted, low-cost supplemental field action is recommended to fully close the ESM gap before the planned March 2026 validation audit.

Part 1 — Bulk Density Sampling at t=0

1.1 What the Methodology Requires

VM0042 v2.2 Section 8.2.1.5 requires that bulk density be determined using the core, excavation, or clod method and that estimates be “*derived from representative (unbiased) sampling*” with accuracy ensured “*through adherence to best practices*.”¹ The methodology does not prescribe a fixed number of BD measurements per SOC sample point, nor does it require one BD measurement per SOC sample. The operative requirement is statistical representativeness at the stratum level.

Section 8.2.1.3 further states that SOC content and soil mass should be obtained from the same sample “*where possible*” — language that explicitly acknowledges practical constraints and permits a subsample approach where full BD coverage is not feasible.

1.2 Statistical Justification for Stratum-Average BD

Where BD is measured on a representative subsample and variance within the stratum is demonstrably low, applying a stratum-average BD to all sample points is statistically defensible. The coefficient of variation (CV) is the standard metric used to characterise within-stratum BD homogeneity.² The CV is defined as:

$$CV (\%) = (\sigma / \bar{x}) \times 100$$

where σ is the standard deviation of BD measurements within the stratum and $\bar{x}$ is the stratum mean BD. CVs below approximately 10–15% indicate low spatial variability and are widely accepted as justification for applying a stratum mean.³

1.3 Results: BD Variability in Project Strata

Bulk density was measured on a randomly selected subset of samples within each soil stratum during the September 2025 field campaign. Table 1 summarises BD sample counts and variability statistics by stratum, excluding the Adu cooperative which is not participating in this project.

Soil Stratum (FAO Code)	Cooperatives	Total Samples	BD Samples (0–30 cm)	BD Rate	CV (%)
ANm	Eor Emayian, Lanyuak	61	14	23%	6.8%
NTu	Kabianga, Kipsigis, Mumberes	100	16	16%	8.7%
NTh	Mumberes	13	6	46%	5.7%
Total	5 cooperatives	174	36	21%	—

Table 1. Bulk density sampling summary by soil stratum (non-Adu cooperatives). CV calculated from measured BD subsamples within each stratum. Total BD samples reflect the actual dataset ($n=36$).

All three strata return CVs well below the 10–15% threshold, ranging from **5.7% (NTh)** to **8.7% (NTu)**. BD measurements within the multi-cooperative NTu and ANm strata are distributed across all constituent cooperatives (NTu: Kabianga 6, Kipsigis 4, Mumberes 6; ANm: Eor Emayian 8, Lanyuak 6), confirming spatial representativeness.

Key Finding — Part 1

The project does **not** need to return to the field to collect BD measurements on every SOC sample point. The existing BD subsample, randomly distributed across strata with CVs of 5.7–8.7%, is statistically sufficient to justify stratum-average BD application and is fully consistent with VM0042 v2.2 requirements.

For NTh specifically, the 6 existing measurements exceed the stratum-specific minimum sample size of $n=5$ derived from its own CV (see Section 2.4 for the per-stratum sample size analysis).

¹ Verra (2023). VM0042 Improved Agricultural Land Management, v2.2. Washington DC: Verra. Sections 8.2.1.3 and 8.2.1.5.

² De Gruijter, J. et al. (2006). Sampling for Natural Resource Monitoring. Springer, Berlin. CV is the recommended metric for characterising within-stratum homogeneity of soil physical properties.

³ FAO (2020). Measuring and modelling soil carbon stocks and stock changes in livestock production systems. FAO, Rome. CV thresholds for bulk density stratification are discussed in Chapter 4.

Part 2 — Equivalent Soil Mass (ESM): Requirements, Timing, and Additive Sampling

2.1 What is ESM and Why Does it Matter?

Soil organic carbon stocks are conventionally expressed per unit area to a fixed depth (e.g., 0–30 cm). However, a fixed depth contains a *different mass of soil* depending on bulk density. If improved land management practices alter BD over time — for example, reduced tillage may change surface BD —

then comparing fixed-depth SOC stocks at two time points conflates genuine carbon change with a volumetric artefact. ESM corrects for this by expressing SOC stocks relative to a fixed *mass* of soil rather than a fixed depth.⁴

The ESM approach adopted under VM0042 v2.2 follows the Wendt & Hauser (2013) cubic spline method.⁵ The procedure requires a soil mass profile — measurements at two depth increments (0–30 cm and 30–50 cm) — to fit a spline through cumulative soil mass vs. cumulative SOC mass with depth. A single reference soil mass is then defined as the cumulative soil mass to 30 cm of the densest sample in the dataset, and all samples are interpolated to that reference mass for consistent comparison.

2.2 When is ESM Required? — Timing Under Approach 1

Under Quantification Approach 1, VM0042 v2.2 Section 8.2.1.3 states that SOC stock changes from direct measurements must be reported on an ESM basis. This requirement applies to the **comparison of SOC stocks across two time periods** — i.e., at model true-up events. The methodology (Table 8) requires remeasurement “every five years or more frequently” under Approach 1.⁶

At baseline (t=0), soil samples serve a different and more limited purpose: they are **model initialisation inputs** for RothC, requiring SOC content (%) and BD to calculate initial SOC stocks via Equation (4) of VM0042 v2.2:

$$\text{SOC}_{\text{model}} = 100 \times \text{BD}_{\text{corr}} \times d \times \text{OC}_{n,dl}$$

where BD_{corr} is corrected bulk density of the fine soil fraction (g/cm^3), d is soil depth (cm), and $\text{OC}_{n,dl}$ is organic carbon content in sample n at depth layer dl . This calculation requires only a single depth layer and does not invoke the ESM cubic spline framework.

Key Finding — Part 2a

The project is not required to have collected two depth increments at t=0 for baseline model initialisation. The 0–30 cm composite samples collected in September 2025 are fully adequate as RothC inputs. The two-increment ESM requirement does not apply until the first model true-up, currently anticipated around 2030 (t≈5).

To avoid ambiguity at validation, the monitoring plan should explicitly state that: (i) 0–30 cm composite samples at t=0 are used solely for model initialisation per VM0042 v2.2 Table 6; and (ii) two-increment sampling for ESM will be conducted at the first true-up event per Section 8.2.1.3 and Wendt & Hauser (2013).

2.3 Can the 30–50 cm BD Collection Occur Separately from the t=0 SOC Sampling?

A key practical question is whether the supplemental 30–50 cm BD cores required for ESM baseline characterisation (see Section 2.5) may be collected at a different time than the original September 2025 SOC sampling campaign. The answer is **yes**, subject to the conditions set out below.

VM0042 v2.2 Section 8.2.1.3 uses the qualifier “*where possible*” when specifying that SOC content and soil mass should be obtained from the same sample.¹ This language explicitly acknowledges operational and logistical constraints and does not impose a strict co-sampling requirement. The 30–50 cm BD cores are not collected for SOC analysis — they are needed only to characterise the subsoil BD profile for the ESM soil mass computation. Because subsoil BD (below the tillage zone, >25 cm) is a relatively stable physical property that does not respond rapidly to management changes,⁷ a short time lag between the SOC sampling event and the supplemental BD collection does not materially compromise the baseline soil mass profile.

The following conditions must be met to ensure the separate-timing approach is defensible at validation:

- **Spatial co-location:** The 30–50 cm BD cores must be taken within 1–2 m of the GPS-georeferenced t=0 SOC sample points, so that the two depth increments are spatially paired for the ESM profile.
- **Minimal time lag:** The supplemental cores should be collected as soon as practicable and no later than the project validation event (planned March 2026). A gap of a few months is unlikely to alter subsoil BD materially; any gap exceeding one growing season should be documented and justified.
- **Documentation:** The monitoring plan and project description should explicitly state the timing and rationale for the separate BD collection, the GPS locations used, and the basis for concluding that subsoil BD has not changed materially between the two collection events.
- **No SOC re-analysis required:** The 30–50 cm cores are for BD measurement only. No laboratory SOC analysis is required, keeping the supplemental campaign low-cost.

Why seasonality doesn't affect bulk density:

Critically, bulk density is defined using the oven-dry mass of soil, meaning that all moisture is removed prior to measurement. As such, it represents only the mass of soil solids relative to the total volume, excluding any contribution from water. This standardization ensures that bulk density is an intrinsic physical property of the soil's structure and packing, rather than a function of its transient moisture content. Although soil water content may influence field sampling conditions or cause minor volume changes in shrink–swell soils, these effects are methodological rather than conceptual. When properly measured (i.e., using undisturbed cores and oven-drying), bulk density remains independent of soil moisture. Therefore, comparing bulk density measurements across seasons — such as between wet and dry periods in the western highlands of Kenya — is scientifically valid, provided that consistent sampling and drying protocols are followed. Seasonal differences in moisture do not directly affect bulk density values, allowing for meaningful comparison of soil structural conditions over time.

Key Finding — Part 2b (Timing)

VM0042 v2.2 does not prohibit separate-timing BD collection for ESM baseline characterisation. The 30–50 cm BD cores required for ESM at t=5 may be collected in a dedicated low-cost field campaign prior to validation, provided they are spatially co-located with the t=0 SOC sample points (within 1–2 m GPS), the time lag is documented and justified, and the monitoring plan explicitly describes this approach.

No SOC laboratory analysis is required on the additional cores.

2.4 Stratum-Specific Minimum Sample Sizes for 30–50 cm BD Sampling

VM0042 v2.2 does not prescribe a specific sample count for supplemental BD characterisation. The appropriate number is determined by the precision required to estimate stratum-mean BD at 30–50 cm. Since subsoil BD is a relatively stable physical property below the tillage zone, variability is typically lower than at the surface.⁷ We apply the standard precision-target sample size formula recommended in FAO (2019):⁸

$$n \geq (t_{\alpha} \times CV / p)^2 \quad \dots \text{(Equation 1)}$$

Critically, rather than applying a single conservative CV across all strata (as would be the case if the worst-case NTu value of 8.7% were used uniformly), this note applies a **stratum-specific CV** from the actual project dataset for each stratum. This is the more appropriate approach since each stratum

has its own measured BD variability, and applying an inflated CV to the NTh stratum — which is both the lowest-variance and smallest-area stratum — would be unnecessarily conservative.

Soil Stratum (FAO Code)	CV (%) (0–30 cm)	Precision Target (p)	t_{α}	n (calculated)	n (adopted) 0–30 cm BD	n (adopted) 30–50 cm BD
ANm	6.8%	±5%	1.96	8	14	8
NTu	8.7%	±5%	1.96	12	16	12
NTh	5.7%	±5%	1.96	5	6	6

Table 2. Stratum-specific sample size calculations for 30–50 cm BD sampling, using each stratum’s measured CV from the September 2025 campaign. All strata already meet or exceed the calculated minimum with their existing 0–30 cm BD subsamples.

All three strata meet the precision target with their existing 0–30 cm BD subsamples. For NTh specifically, the minimum required sample size is **n=5** (derived from its own CV of 5.7%), and the existing 6 measurements already satisfy this threshold. The same stratum-specific approach determines the number of new 30–50 cm BD cores needed per stratum (Section 2.5).

The per-stratum calculations are as follows:

- **ANm:** $n \geq (1.96 \times 6.8 / 5)^2 = (2.66)^2 \approx 7.1 \rightarrow n = 8$
- **NTu:** $n \geq (1.96 \times 8.7 / 5)^2 = (3.41)^2 \approx 11.6 \rightarrow n = 12$
- **NTh:** $n \geq (1.96 \times 5.7 / 5)^2 = (2.23)^2 \approx 5.0 \rightarrow n = 5$ (6 existing samples satisfy this)

2.5 The ESM Gap at t=5 — and How to Close It

While single-depth sampling at t=0 is acceptable for model initialisation, a legitimate planning question arises for t=5: without a characterised 30–50 cm baseline, can the ESM comparison be performed robustly? The answer is yes, provided the 30–50 cm BD profile is characterised as a supplemental baseline measurement before validation. The deeper layer informs the soil mass profile shape only — no additional SOC analysis is required on the deeper cores.

The project will collect new 30–50 cm BD cores at a representative subset of georeferenced sample points per stratum. The number of cores per stratum is set equal to the stratum-specific minimum n from Equation 1. These cores, combined with the existing 0–30 cm BD values at the same locations, yield complete two-increment soil mass profiles at baseline for the ESM computation at t=5.

Stratum	Cooperatives	Total Samples	0–30 cm BD (existing)	30–50 cm BD (new/additive)	BD per depth	Total BD Measurements
ANm	Eor Emayian, Lanyuak	61	14	8 (new)	8	22
NTu	Kabianga, Kipsigis, Mumberes	100	16	12 (new)	12	28
NTh	Mumberes	13	6	6 (new)	6	12
Total	5 cooperatives	174	36	26	—	58

Table 3. Complete BD sampling plan — existing 0–30 cm measurements plus planned 30–50 cm additive samples. New 30–50 cm core counts are based on stratum-specific minimum sample sizes from Equation 1. All new cores to be collected at or adjacent to existing georeferenced t=0 locations. No SOC analysis required on new cores.

Summary: What the Project Does and Does Not Need to Do

- **Does NOT need to do:** Remeasure BD at all 174 sample points. Existing BD subsamples with CVs of 5.7–8.7% are statistically sufficient for stratum-average application across all three strata.
- **Does NOT need to do:** Re-collect 0–30 cm samples at two depths. Single-depth composites are fully acceptable for RothC model initialisation at t=0.
- **Does NOT need to do:** Conduct ESM analysis now. ESM is required only at the first model true-up (~2030), not at baseline.
- **Does NOT need to do:** Collect 30–50 cm BD cores simultaneously with the original SOC sampling. VM0042 v2.2's "where possible" language permits separate-timing collection, subject to spatial co-location and documentation requirements (Section 2.3).
- **SHOULD do (before validation):** Collect 30–50 cm BD cores using stratum-specific sample sizes (ANm: 8, NTu: 12, NTh: 6 — 26 new cores total) at existing georeferenced points to establish the baseline soil mass profile for ESM at t=5. No SOC analysis required on these cores.

⁴ Ellert, B.H. & Bettany, J.R. (1995). Calculation of organic matter and nutrients stored in soils under contrasting management regimes. *Canadian Journal of Soil Science*, 75(4), 529–538. Foundational ESM paper.

⁵ Wendt, J.W. & Hauser, S. (2013). An equivalent soil mass procedure for monitoring soil organic carbon in multiple soil layers. *European Journal of Soil Science*, 64(1), 58–65. The cubic spline ESM spreadsheet referenced in VM0042 v2.2 is available on the VM0042 Verra webpage.

⁶ VM0042 v2.2, Table 6 and Table 8 govern model input requirements for Quantification Approach 1 at t=0 and at subsequent remeasurement events respectively.

⁷ Grossman, R.B. & Reinsch, T.G. (2002). Bulk density and linear extensibility. In: *Methods of Soil Analysis, Part 4*. Soil Science Society of America, Madison, WI. Subsoil BD (>25 cm) typically exhibits lower spatial variability than surface layers subject to tillage.

⁸ FAO (2019). *Soil Organic Carbon: the hidden potential*. FAO, Rome. Precision-target sampling framework adapted from Chapter 5; the same FAO (2019) reference is cited directly in VM0042 v2.2 Section 8.2.1.3.